



Abnormal post-exercise depression of corticomotor excitability is associated with fatigue in post-COVID-19 condition

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ABSTRACT

Background: Persistent fatigue is a disabling symptom of post-COVID-19 condition (PCC), but objective neurophysiological markers of post-exertional dysfunction remain poorly defined. Post-exercise depression (PED) of corticomotor excitability, assessed by transcranial magnetic stimulation (TMS), is an index of central fatigability, but its relationship with subjective fatigue in PCC is unclear.

Methods: We studied 58 patients with PCC and 22 age- and sex-matched healthy controls (HCs). Participants performed a 1-min sustained maximal voluntary pinching task. Corticomotor excitability was assessed before and after exercise by motor-evoked potential (MEP) amplitude and cortical silent period (SP) duration recorded from the first dorsal interosseous muscle. Fatigue severity and perceived exertion were assessed using the Fatigue Severity Scale, Fatigue Rating Scale, and Borg CR100 scale.

Results: Post-exercise modulation of corticomotor excitability differed between groups. Compared with HCs, PCC patients showed a smaller reduction in MEP amplitude. HCs showed post-exercise SP prolongation, whereas patients showed minimal change or paradoxical shortening. Lower MEP reduction and reduced SP prolongation correlated with higher fatigue and exertion.

Conclusions: PED abnormalities suggest altered regulation of corticomotor output after exertion in PCC.

Significance: PED may provide an objective neurophysiological marker of altered central fatigability.

1. Introduction

Despite substantial progress in managing acute COVID-19, which has now entered an endemic phase, a subset of individuals continues to experience persistent symptoms well beyond the initial infection. This long-lasting clinical picture is commonly referred to as 'Post-COVID-19 condition' (PCC) or 'long COVID' and has recently been included in the 11th revision of the International Classification of Diseases (Nasserie et al., 2021, National Academies of Sciences et al., 2024, Soriano et al., 2022).

One of the most common and disabling complaints of PCC is persistent fatigue (Carfi et al., 2020). Fatigue is defined as the inability to sustain or even begin physical activity, accompanied by an

overwhelming feeling of exhaustion that does not improve with rest. In neurological disorders, fatigue may result from metabolic or structural brain lesions that disrupt the normal activation process of interconnected pathways among the basal ganglia, thalamus, limbic system, and higher motor and pre-motor centers (Chaudhuri and Behan, 2004).

In clinical settings, fatigue is usually captured as a subjective experience reported by the patient, whereas fatigability refers to measurable changes in performance that emerge during or after sustained activity. For this reason, the two constructs are related but not interchangeable, and they require partly different assessment strategies (Tankisi et al., 2024).

A reliable neurophysiological approach for investigating central components of fatigability is to assess post-exercise depression (PED),

Abbreviations: PCC, post-COVID-19 condition; HCs, healthy controls; TMS, transcranial magnetic stimulation; MCV, maximum voluntary contraction; PED, post-exercise depression; MEP, motor evoked potentials; SP, silent period; FSS, Fatigue Severity Scale, FRS, Fatigue Rating Scale; Borg Scale, Borg Category Ratio (CR)100 Scale; LICl, long-interval intracortical inhibition.

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